

RFID and Motion-Based Attendance System (Smart Rollcall)

Components Used:

1. **ESP32**
2. **LCD Display (I2C)**
3. **RFID Reader (RC522)**
4. **Motion Detection Sensor Module**
5. **Buzzer**
6. **Green and Red LED Lights with Resistors**
7. **Jumper Wires**
8. **RTC Module HW-111**

Project Overview:

This project aims to provide an efficient, secure, and easy-to-use attendance system. The **ESP32** microcontroller is used for two key functions:

1. **Control** the system.
2. **Wi-Fi** connectivity to send attendance data online to an Excel sheet.

Components and Functionality:

- **RFID Reader:** Used to read NFC cards and key fobs for marking attendance.
- **Motion Detection Sensor:** Ensures attendance is securely recorded. If someone taps their NFC card and passes through, motion must be detected to mark attendance. The **green LED** will flash and the **buzzer** will beep once.
 - If no motion is detected, the **red LED** will flash, and the **buzzer** will beep longer.
- **RTC Module:** Provides accurate time for logging attendance in the Google Sheet.

Key Features:

1. **Motion Detection for Secure Attendance:**
 - Attendance is only recorded if motion is detected after tapping the NFC card.
2. **Master Card Lock:**
 - A master NFC card locks/unlocks the system to control when attendance marking is possible.
3. **Real-Time Logging:**
 - Attendance data (UID, status, timestamp) is sent to Google Sheets via Wi-Fi.
4. **Visual & Audio Feedback:**
 - **Green LED + single beep:** Attendance marked.
 - **Red LED + long beep:** Attendance not marked (invalid card or no motion detected).
5. **RTC for Accurate Timestamps:**

- Ensures precise time logging for the Google Sheet.

Workflow:

1. System Initialization:

- ESP32 connects to Wi-Fi.
- LCD displays "System Ready".

2. Master Card Tap:

- Tap the master card to enable or disable the attendance system.
- LCD shows "System Locked/Unlocked".

3. Attendance Marking:

- The user taps their NFC card.
- If motion is detected within 10 seconds:
 - The **green LED** flashes, and the **buzzer** beeps once.
 - Attendance is logged to Google Sheets.
- If no motion is detected:
 - The **red LED** flashes, and the **buzzer** beeps longer.
 - Attendance is not logged.

4. Real-Time Logging:

- Data (UID, status, timestamp) is sent to Google Sheets via Wi-Fi.

Connections:

Component	ESP32 Pin	Notes
RFID RC522 (SDA)	GPIO 5	SPI SS/CS pin
RFID SCK	GPIO 18	SPI SCK
RFID MOSI	GPIO 23	SPI MOSI
RFID MISO	GPIO 19	SPI MISO
RFID RST	GPIO 22	
LCD I2C (SDA)	GPIO 21	I2C Data
LCD I2C (SCL)	GPIO 22	I2C Clock (shared with RFID RST if needed)
PIR Sensor (OUT)	GPIO 4	Motion detection signal
Buzzer	GPIO 15	Active buzzer (PWM optional)

Component	ESP32 Pin	Notes
Green LED	GPIO 12	With 220Ω resistor
Red LED	GPIO 13	With 220Ω resistor
RTC Module (SDA)	GPIO 21	Shared I2C with LCD
RTC Module (SCL)	GPIO 22	Shared I2C with LCD
Power		Use ESP32's 3.3V for RFID, 5V for PIR/LCD

Code Implementation:

```
#include <SPI.h>

#include <MFRC522.h>

#include <WiFi.h>

#include <HTTPClient.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <DS3231.h>


// Wi-Fi Credentials

const char* ssid = "YOUR_WIFI_SSID";

const char* password = "YOUR_WIFI_PASSWORD";


// Google Apps Script URL

const String GOOGLE_SCRIPT_URL = "https://script.google.com/macros/s/XYZ789/exec";


// Pins

#define SS_PIN 5 // RFID SDA

#define RST_PIN 22 // RFID RST

#define PIR_PIN 4 // Motion Sensor

#define BUZZER 15 // Buzzer

#define GREEN_LED 12 // Green LED

#define RED_LED 13 // Red LED


// Objects

MFRC522 rfid(SS_PIN, RST_PIN);

LiquidCrystal_I2C lcd(0x27, 16, 2); // I2C address 0x27, 16x2 LCD
```

DS3231 rtc;

// Master Card UID (replace with your master card's UID)

byte masterUID[] = {0x12, 0x34, 0x56, 0x78};

bool systemLocked = false;

void setup() {

 Serial.begin(115200);

 SPI.begin();

 Wire.begin();

 // Initialize RFID

 rfid.PCD_Init();

 // Initialize LCD

 lcd.init();

 lcd.backlight();

 lcd.print("System Ready!");

 // Initialize RTC

 // rtc.setDateTime(DATE , TIME);

 // Configure pins

 pinMode(PIR_PIN, INPUT);

 pinMode(BUZZER, OUTPUT);

 pinMode(GREEN_LED, OUTPUT);

 pinMode(RED_LED, OUTPUT);

 // Connect to Wi-Fi

 WiFi.begin(ssid, password);

 while (WiFi.status() != WL_CONNECTED) {

 delay(500);

 Serial.print(".");

 }

 Serial.println("\nWiFi connected!");

```
lcd.clear();  
  
lcd.print("WiFi Connected!");  
  
}
```

```
void loop() {  
  
  // Check for master card tap  
  
  if (rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial()) {  
  
    if (isMasterCard()) {  
  
      systemLocked = !systemLocked;  
  
      lcd.clear();  
  
      lcd.print(systemLocked ? "System Locked!" : "System Unlocked!");  
  
      delay(2000);  
  
    }  
  
    rfid.PICC_HaltA();  
  
  }  
  
}
```

```
// Attendance marking logic
```

```
if (!systemLocked && rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial()) {  
  
  lcd.clear();  
  
  lcd.print("Card Detected!");  
  
  delay(1000);  
  
}
```

```
// Wait for motion detection
```

```
unsigned long startTime = millis();  
  
bool motionDetected = false;  
  
while (millis() - startTime < 10000) { // 10-second window  
  
  if (digitalRead(PIR_PIN)) {  
  
    motionDetected = true;  
  
    break;  
  
  }  
  
}
```

```
if (motionDetected) {  
  
  lcd.clear();  
  
  lcd.print("Attendance Marked!");  
  
}
```

```

digitalWrite(GREEN_LED, HIGH);

tone(BUZZER, 1000, 500);

sendToGoogleSheet(rfid.uid.uidByte, "PRESENT", "Attendance marked");

delay(2000);

digitalWrite(GREEN_LED, LOW);

} else {

    lcd.clear();

    lcd.print("No Motion Detected!");

    digitalWrite(REDF_LED, HIGH);

    tone(BUZZER, 300, 2000);

    sendToGoogleSheet(rfid.uid.uidByte, "ABSENT", "No motion detected");

    delay(2000);

    digitalWrite(REDF_LED, LOW);

}

rfid.PICC_HaltA();

}

delay(500);

}

```

// Check if scanned card is the master card

```

bool isMasterCard() {

    for (byte i = 0; i < rfid.uid.size; i++) {

        if (rfid.uid.uidByte[i] != masterUID[i]) {

            return false;

        }

    }

    return true;

}

```

// Send data to Google Sheets

```

void sendToGoogleSheet(byte *uid, String status, String remarks) {

    if (WiFi.status() == WL_CONNECTED) {

        HTTPClient http;

        String url = GOOGLE_SCRIPT_URL;
    }
}

```

```
// Convert UID to HEX string

String uidStr;

for (byte i = 0; i < rfid.uid.size; i++) {

    uidStr += String(uid[i], HEX);

}


// Prepare JSON payload

String payload = "{\"uid\":\"" + uidStr + "\", \"status\":\"" + status + "\", \"remarks\":\"" + remarks + "\"}";


http.begin(url);

http.addHeader("Content-Type", "application/json");


int responseCode = http.POST(payload);

if (responseCode == 200) {

    Serial.println("Data sent to Google Sheets!");

} else {

    Serial.println("Error: " + String(responseCode));

}

http.end();

}

}
```